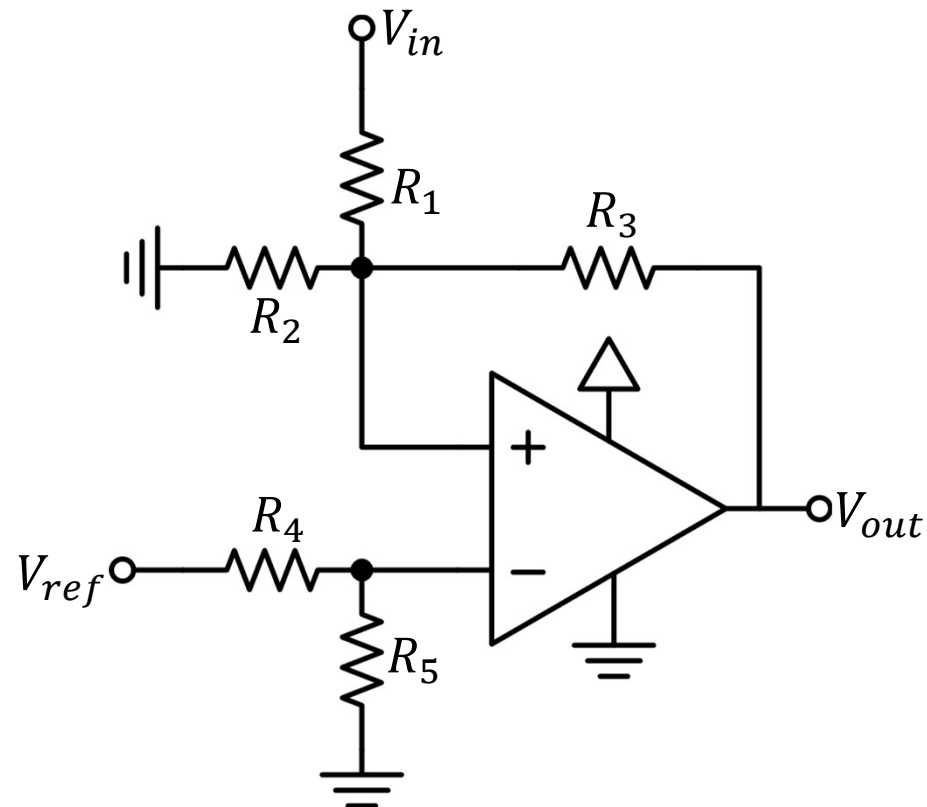


Non-Inverting Comparator



Solution Non-inverting

$$y = 1, x < 1$$

$$x = \frac{V_{ref} V_{DD}}{V_{inH} V_{DD} - V_{ref} V_{inH} + V_{ref} V_{inL}}$$

$$z = \frac{y V_{ref}}{x V_{inH}}$$

- R_3 : choice

$$R_1 = \frac{1 - z}{xz} R_3$$

$$R_2 = \frac{x}{1 - x} R_1$$

Normalise, tolerance, etc

$$x = \frac{R_2}{R_1 + R_2}$$

$$z = \frac{R_3}{xR_1 + R_3}$$

$$V_{inH} = \frac{y V_{ref}}{xz}$$

$$V_{inL} = \frac{y V_{ref} - (1 - z) V_{DD}}{xz}$$

Solution Non-inverting

$$x = 1, y < 1$$

$$y = \frac{V_{inH} V_{DD}}{V_{ref} V_{DD} + V_{ref} V_{inH} - V_{ref} V_{inL}}$$

$$z = \frac{y V_{ref}}{x V_{inH}}$$

- R_3 : choice

$$R_1 = \frac{1 - z}{z} R_3$$

- R_4 : choice

$$R_5 = \frac{y}{1 - y} R_4$$

Normalise, tolerance, etc

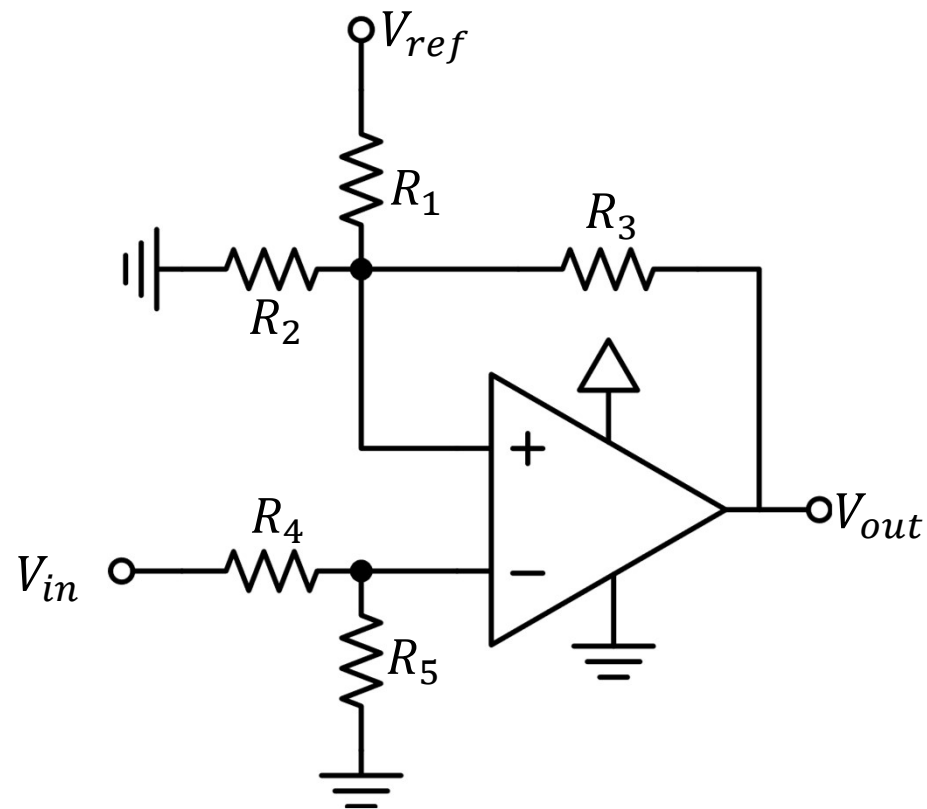
$$y = \frac{R_5}{R_4 + R_5}$$

$$z = \frac{R_3}{x R_1 + R_3}$$

$$V_{inH} = \frac{y V_{ref}}{x z}$$

$$V_{inL} = \frac{y V_{ref} - (1 - z) V_{DD}}{x z}$$

Inverting Comparator



Solution Inverting

$$y = 1, x < 1$$

$$x = \frac{V_{inL} V_{DD}}{V_{inL} V_{ref} + V_{ref} V_{DD} - V_{inH} V_{ref}}$$

$$z = \frac{y V_{inL}}{x V_{ref}}$$

- R_3 : choice

$$R_1 = \frac{1 - z}{xz} R_3$$

$$R_2 = \frac{x}{1 - x} R_1$$

Normalise, tolerance, etc

$$x = \frac{R_2}{R_1 + R_2}$$

$$z = \frac{R_3}{x R_1 + R_3}$$

$$V_{inL} = \frac{xz}{y} V_{ref}$$

$$V_{inH} = \frac{xz V_{ref} + (1 - z) V_{DD}}{y}$$

Solution Inverting

$$x = 1, y < 1$$

$$y = \frac{V_{ref} V_{DD}}{V_{inH} V_{ref} + V_{inL} V_{DD} - V_{inL} V_{ref}}$$

$$z = \frac{y V_{inL}}{x V_{ref}}$$

- R_3 : choice

$$R_1 = \frac{1 - z}{z} R_3$$

- R_4 : choice

$$R_5 = \frac{y}{1 - y} R_4$$

Normalise, tolerance, etc

$$y = \frac{R_5}{R_4 + R_5}$$

$$z = \frac{R_3}{x R_1 + R_3}$$

$$V_{inL} = \frac{xz}{y} V_{ref}$$

$$V_{inH} = \frac{xz V_{ref} + (1 - z) V_{DD}}{y}$$